

Performance Report

Evaluation

Installation	Equipment	Type	No.
APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356

Readings Included

	Date	Voyage / title	Remarks
1	2026-01-24	04FMNW1MA	

Main indications (all Readings)

Reading	1
Description	Most recent
Fuel oil saving potential	●
Light running	●
Fuel index ECU	●
Firing press. AVG	●
Compression press. AVG	●
Scav. air press. RECEIVER	●
Exh. gas temp. TC in AVG	●
Exh. gas temp. TC out AVG	●
TC speed AVG	●
Spec. FO consumption	●

Indications (for Reading 1)

Description	Indication	Comments / recommendations
MIP deviation CYL 1	⚠ High	MOP: load parameters [offset high or low load]. / Indicator cock or measuring device problem. / TDC detection or evaluation. / Unstable load.
TC Evaluation	⚠ Low	Limited accuracy due to missing inputs

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Optimization potential

Influenced by	Quantification		Proposed actions to achieve the saving / reduction
	Fuel oil equiv.	Thermal load	
Firing pressure by injection	0.0 g/kWh	0 °C	
Firing pressure by compression ratio	0.0 g/kWh	0 °C	
Firing pressure by scavenge air pressure	0.0 g/kWh	0 °C	
ENGINE ROOM ventilation	0.0 g/kWh	0 °C	
Suction press. air TC in	0.0 g/kWh	0 °C	
Press. drop accross SAC	0.0 g/kWh	0 °C	
Scav. air : water temp. SAC in	0.3 g/kWh	1 °C	Reduce SAC water inlet temperature to 29.5 °C to reduce scav. air temp. by 2 °C.
Scav. air : temp. RECEIVER	0.2 g/kWh	1 °C	Increase water flow, clean SAC water- and/or air side to reduce scav. air temp. by 2 °C. Target scav. air temp. is 33 °C (incl. reduction of 2 °C by lower SAC cooling water inlet temp. setpoint).*
Press. drop accross ENGINE	0.0 g/kWh	0 °C	
Exh. gas press. TC out	0.0 g/kWh	0 °C	
Light running	0.0 g/kWh	-	
Fuel oil visc. ENGINE in	0.0 g/kWh	0 °C	
	Fuel oil equiv.	Thermal load	Cyl. lub. oil
TOTAL (specific)	0.5 g/kWh	-	
TOTAL (of current)	0.3 %	-	
			CO2 (emission)
TOTAL optimization potential	0.26 t/d	3 °C	0.8 t/d
			SOx (emission)
			3 kg/d
	Cost		
Fuel oil price			
TOTAL saving potential			

Consumption

Measured / calculated values			
Indication	Fuel oil		Cylinder lub oil
Specific consumption reference	168.2 g/kWh		0.70 g/kWh
Specific consumption ISO / MEAS	176.5 g/kWh		0.86 g/kWh
Consumption per hour day *	3946 kg/h	94.7 t/d	18.5 kg/h
Consumption nautical *	210 kg/nmi		
* Measured values, uncorrected			

Injection timing and compression ratio adjustment

Cylinders												
Settings		1	2	3	4	5	6	7	8	9	10	
Proposed adjustments to reach target firing press., positive value means increase of pressure:												
Compr. ratio offset [-]		0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0
pmax offset [bar]		0.1	0.2	-0.1	0.1	-0.2	0.0	0.0	0.1	0.0	0.0	0.0
Common compr. ratio offset [-]	0.9											
Common pmax offset [bar]	-2.5											

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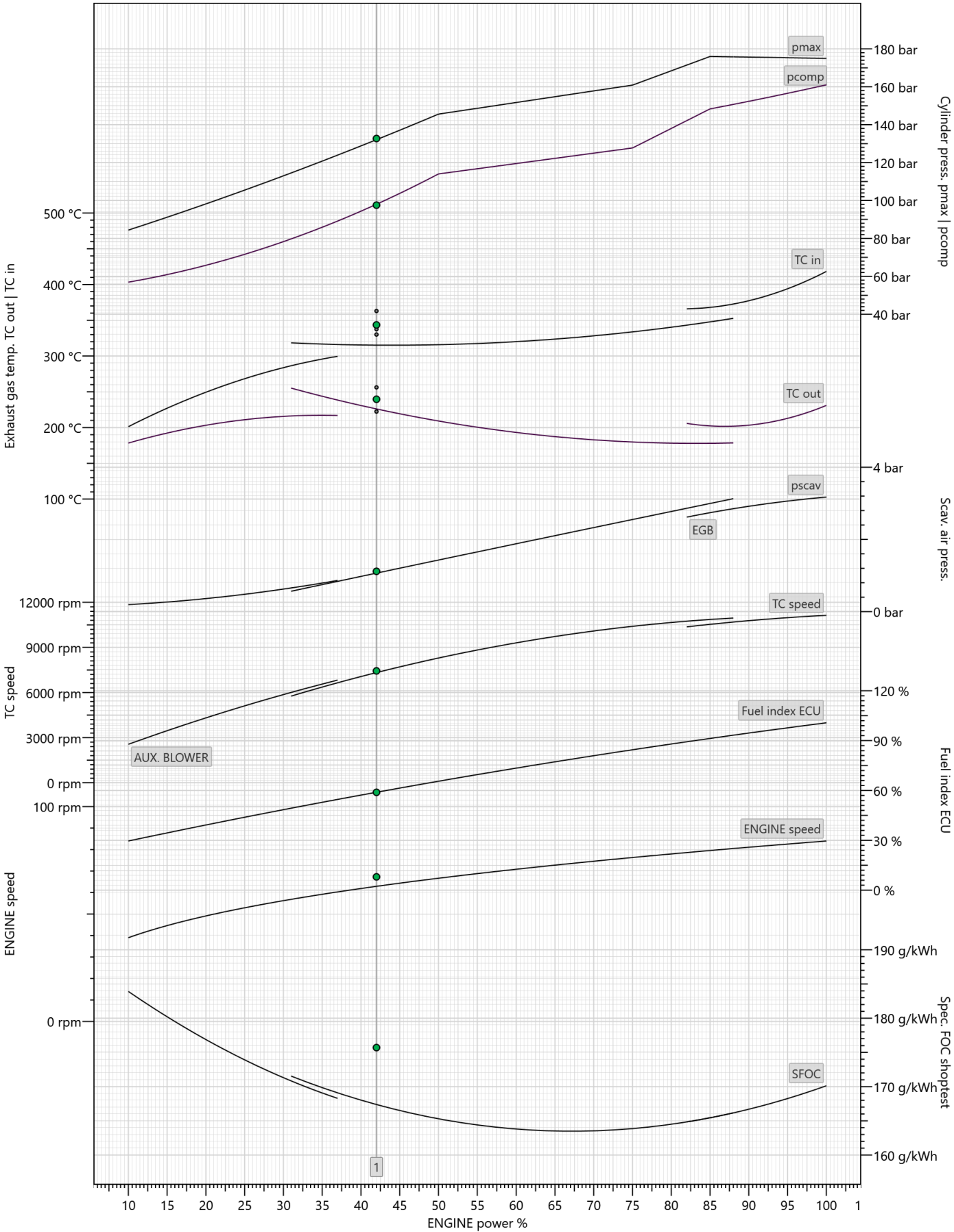
Input plausibility

Value	Comment
Compressor evaluation	
Exhaust gas pressure Manifold	
PTI	Exh. gas press. TC in, higher than expected based on given Press. COMPR. out--> check corresponding readings
PCO	
TTI vs TTO	
TCO	
Minimum required data	
Minimum required data II	
Missing data	
Missing data II	

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Performance Curves

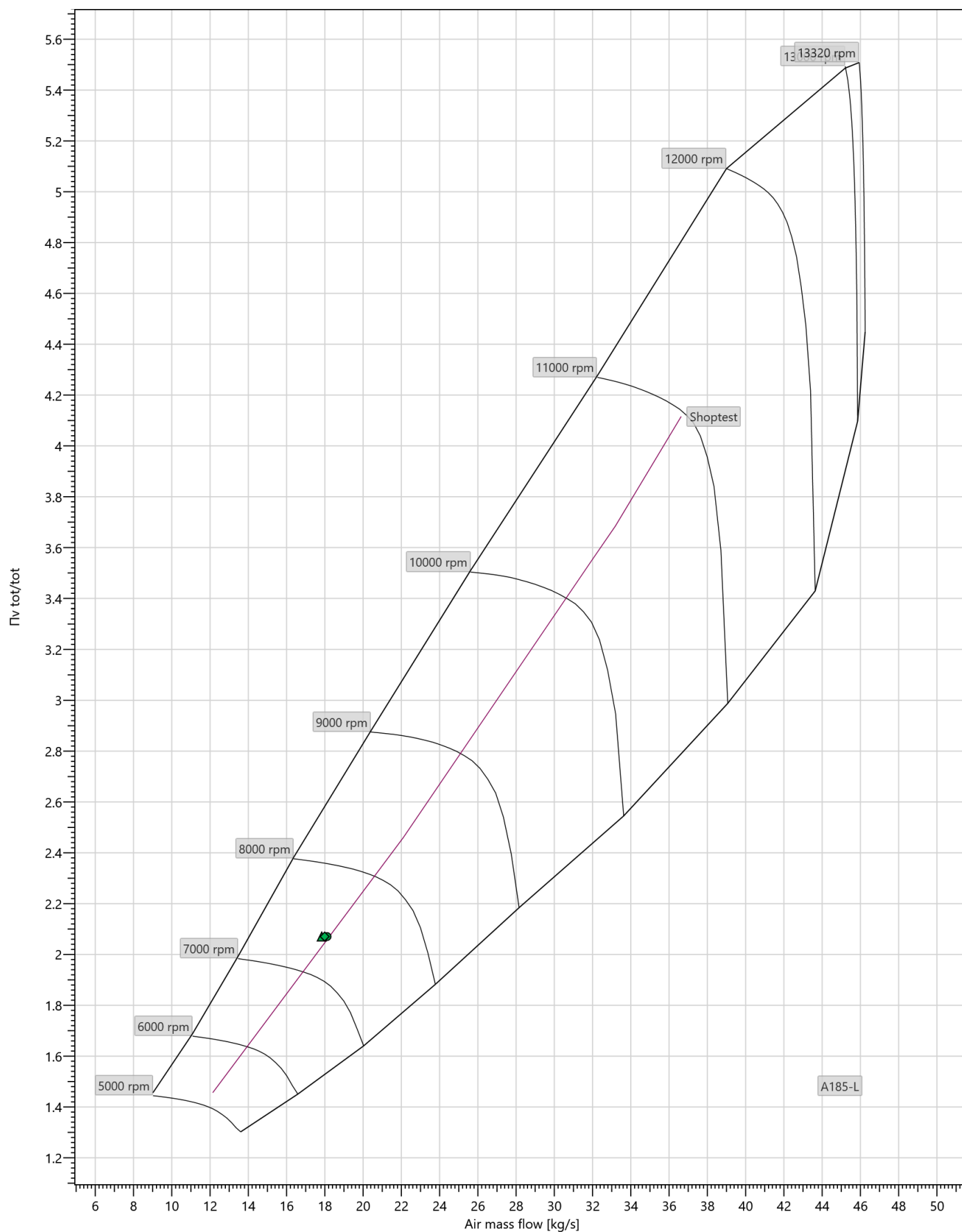
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COMPRESSOR map

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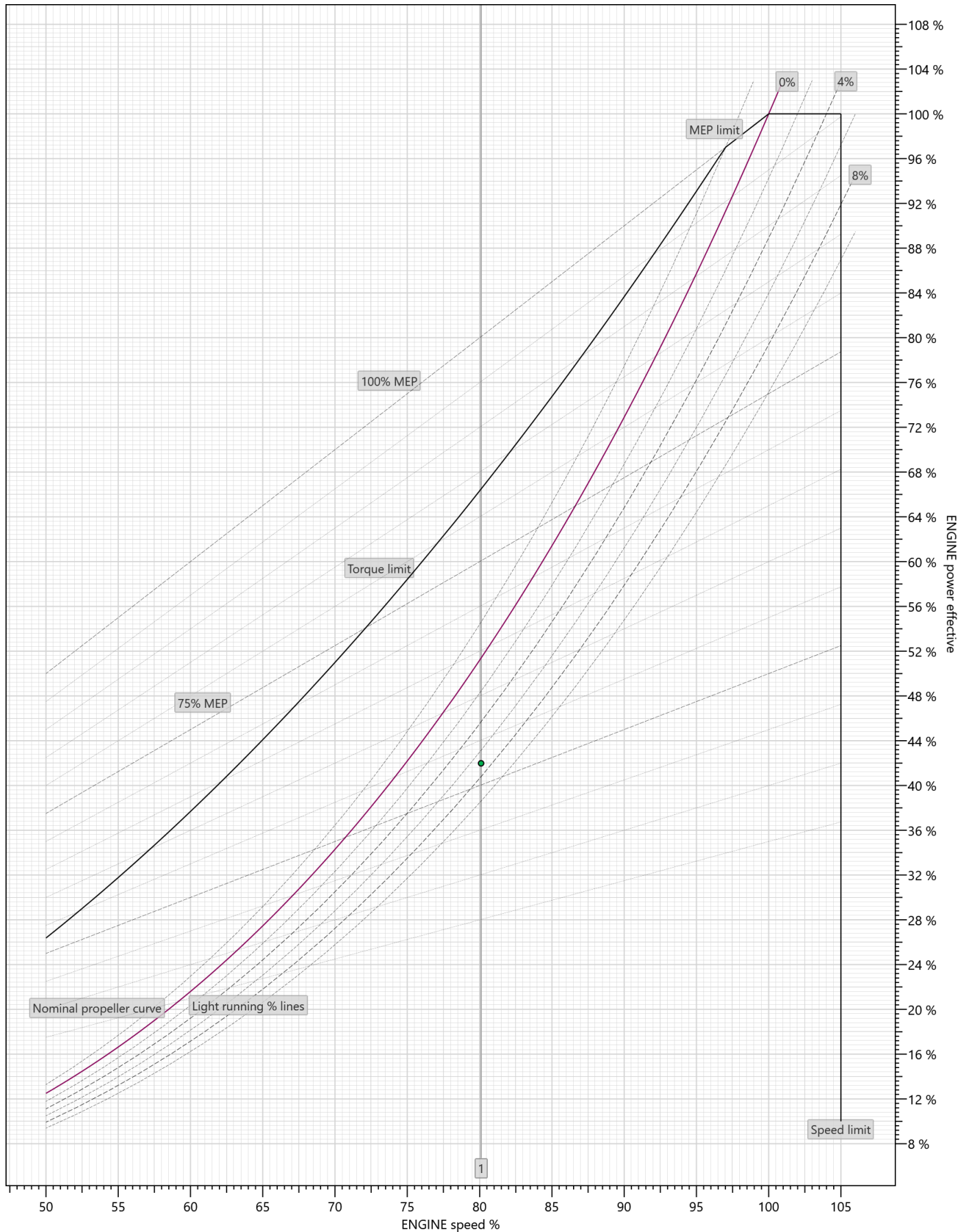


1 2026-01-24 04FMNW1MA		
● TC 1	▲ TC 2	◆ TC 3

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Load Diagram

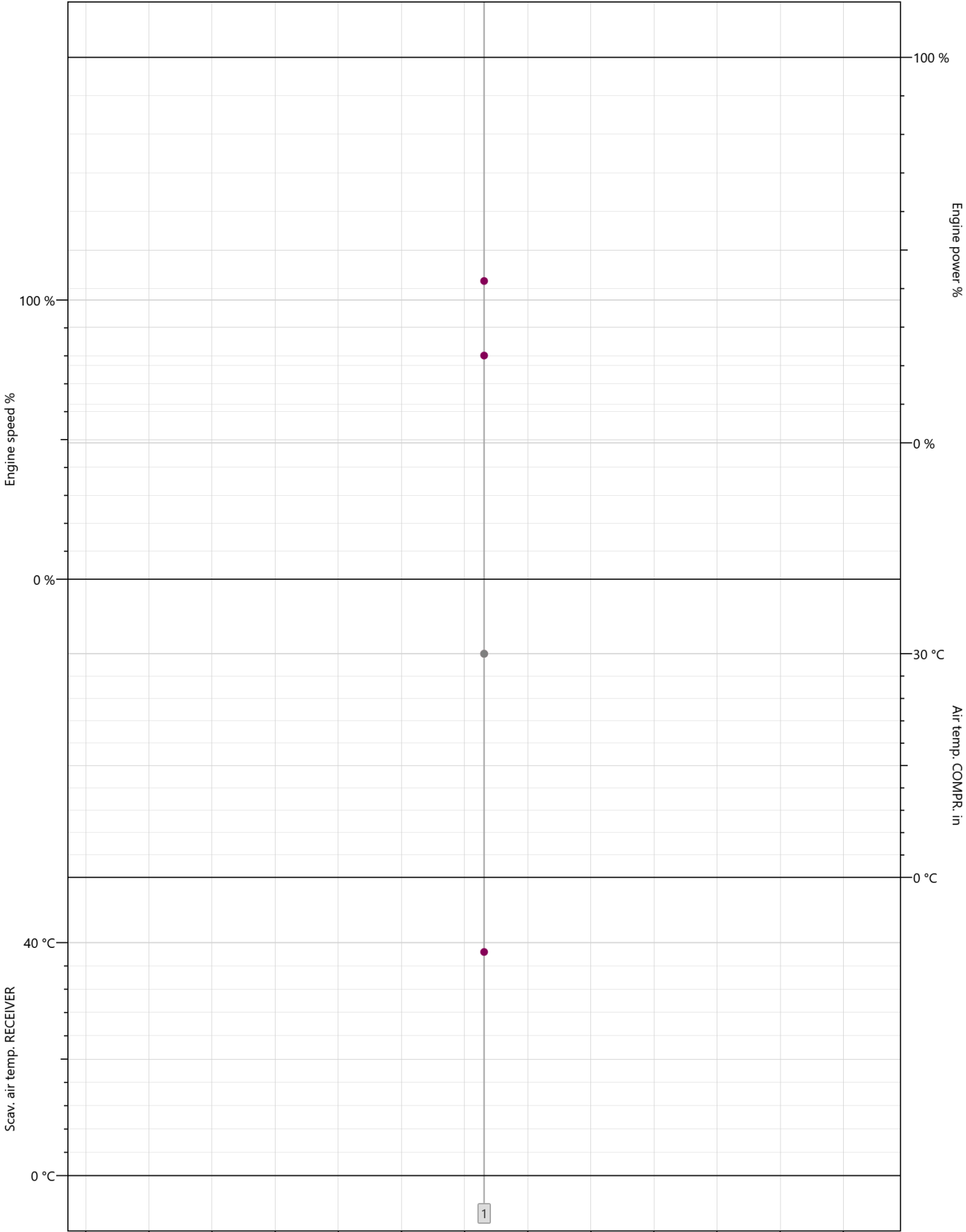
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Trend: LOAD and AMBIENT

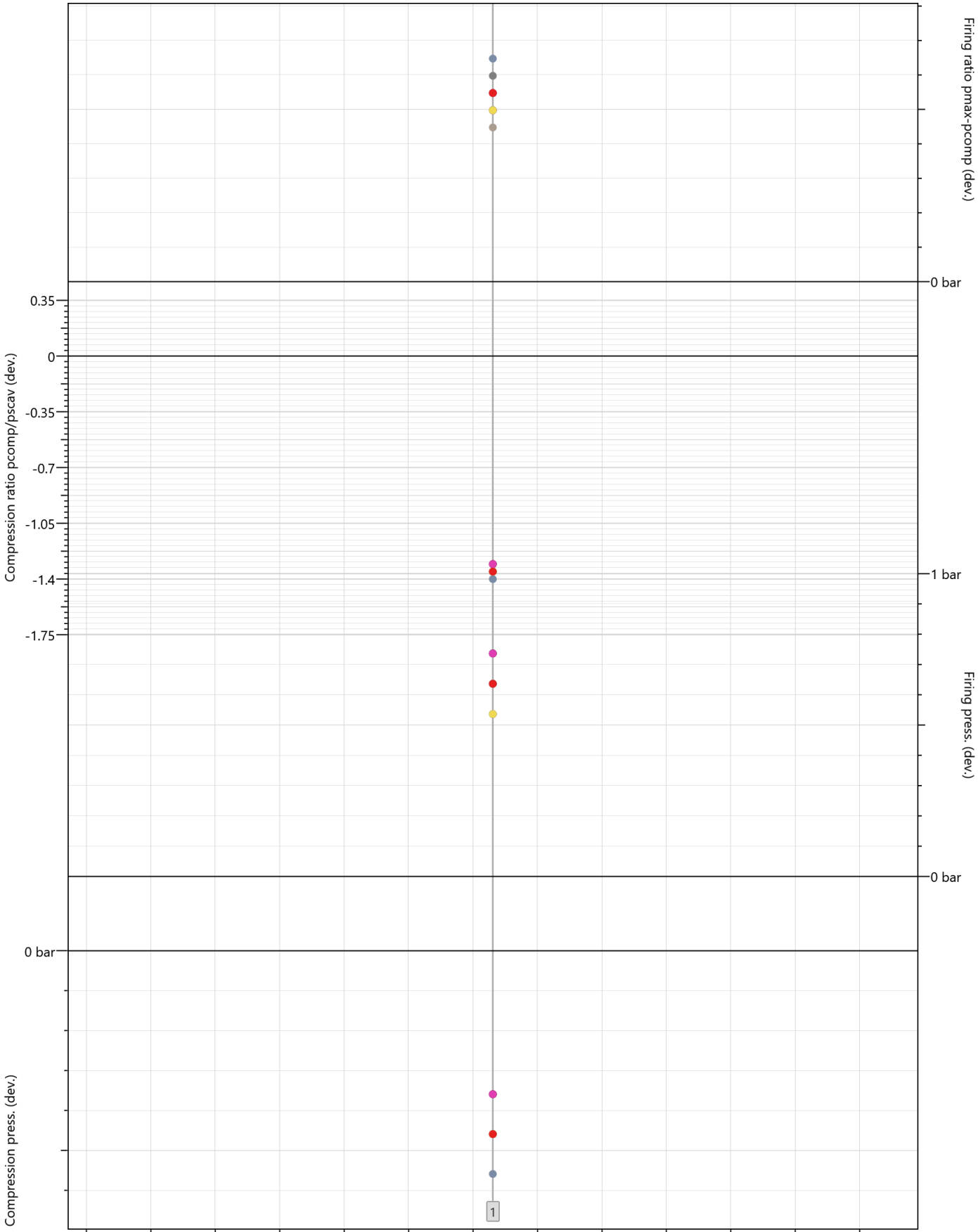
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Trend: COMBUST. CHAMBER

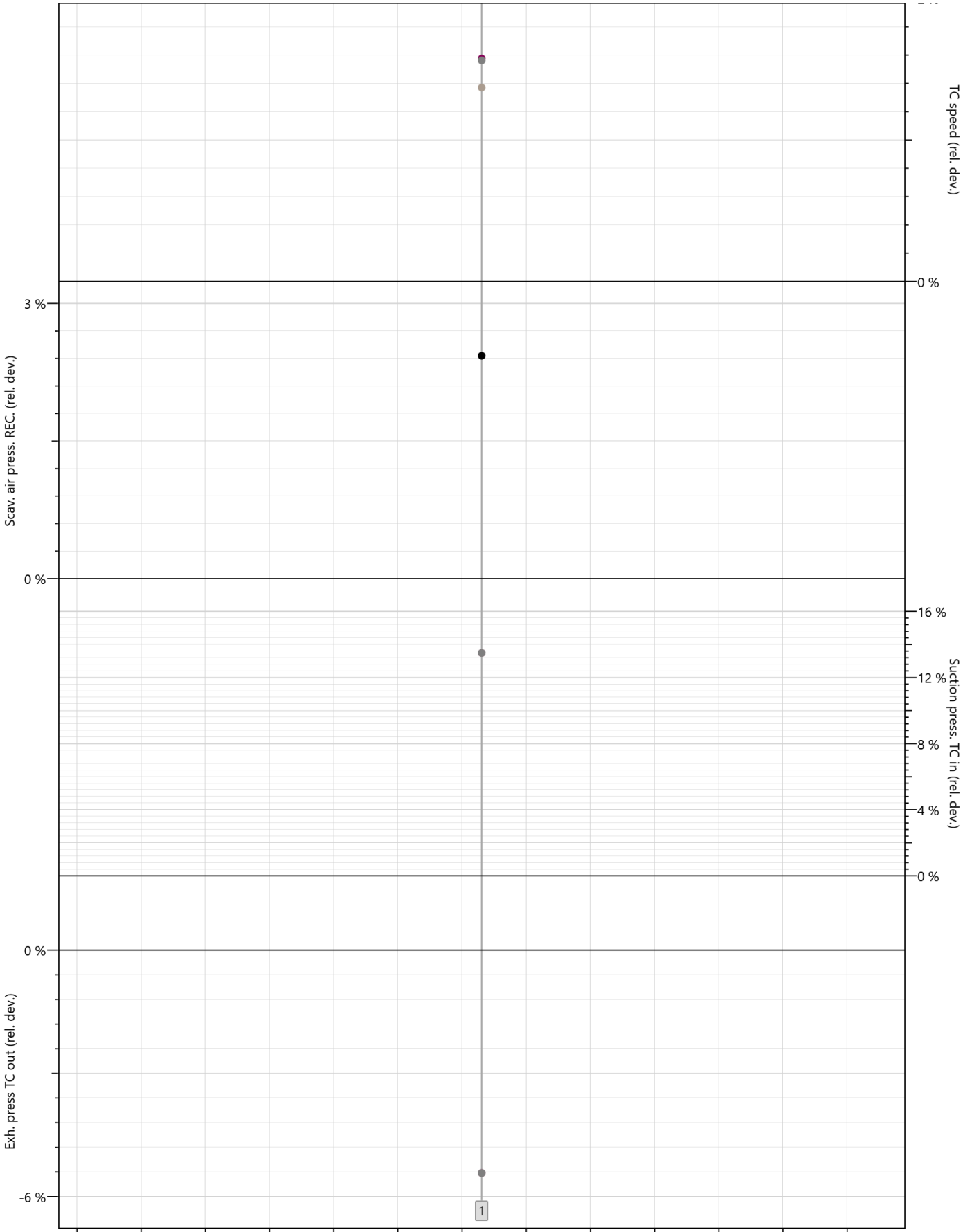
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Trend: TURBOCHARGER

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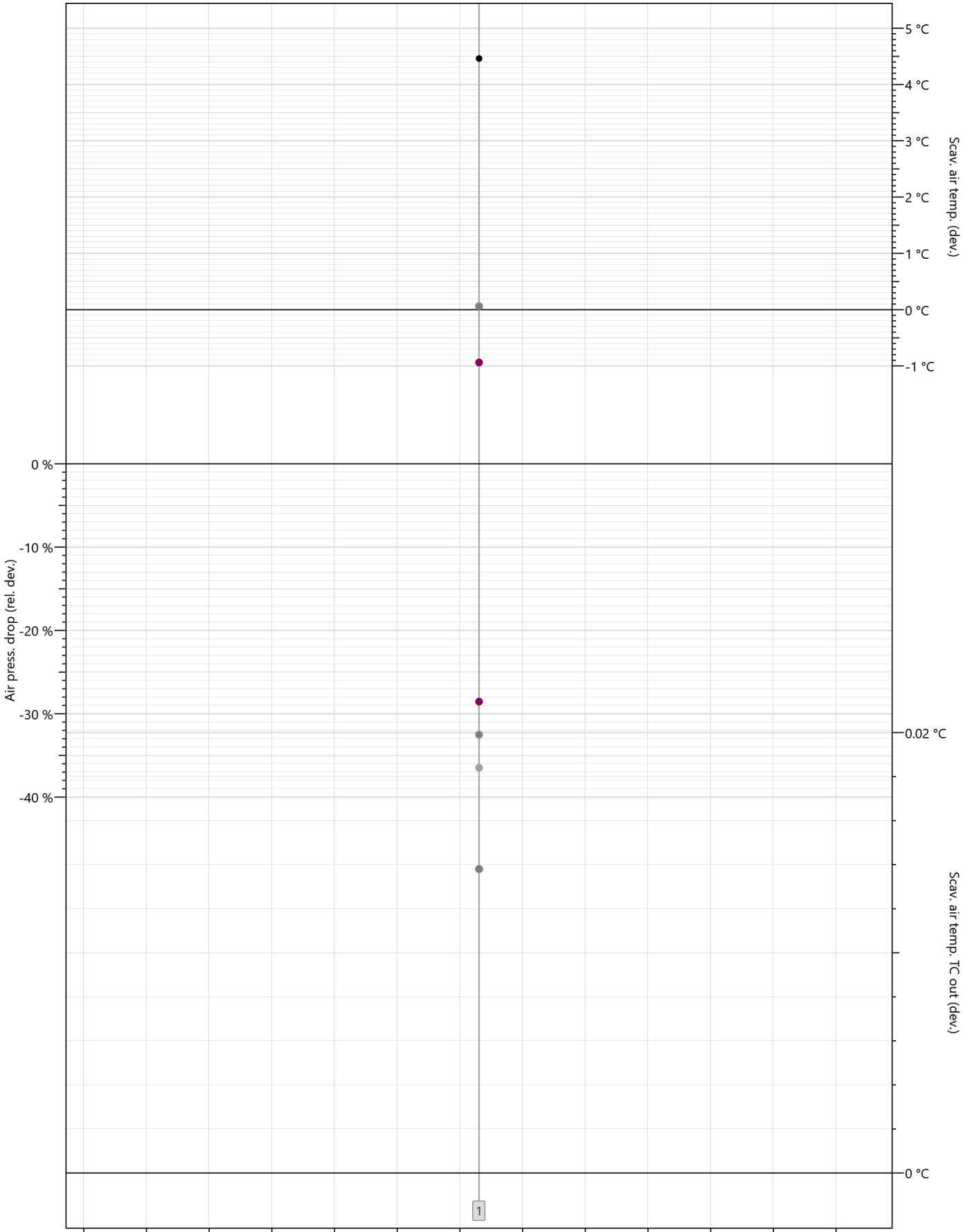


RECEIVER	TC 1	TC 2	TC 3	AVG
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Trend: SCAV. AIR COOLER

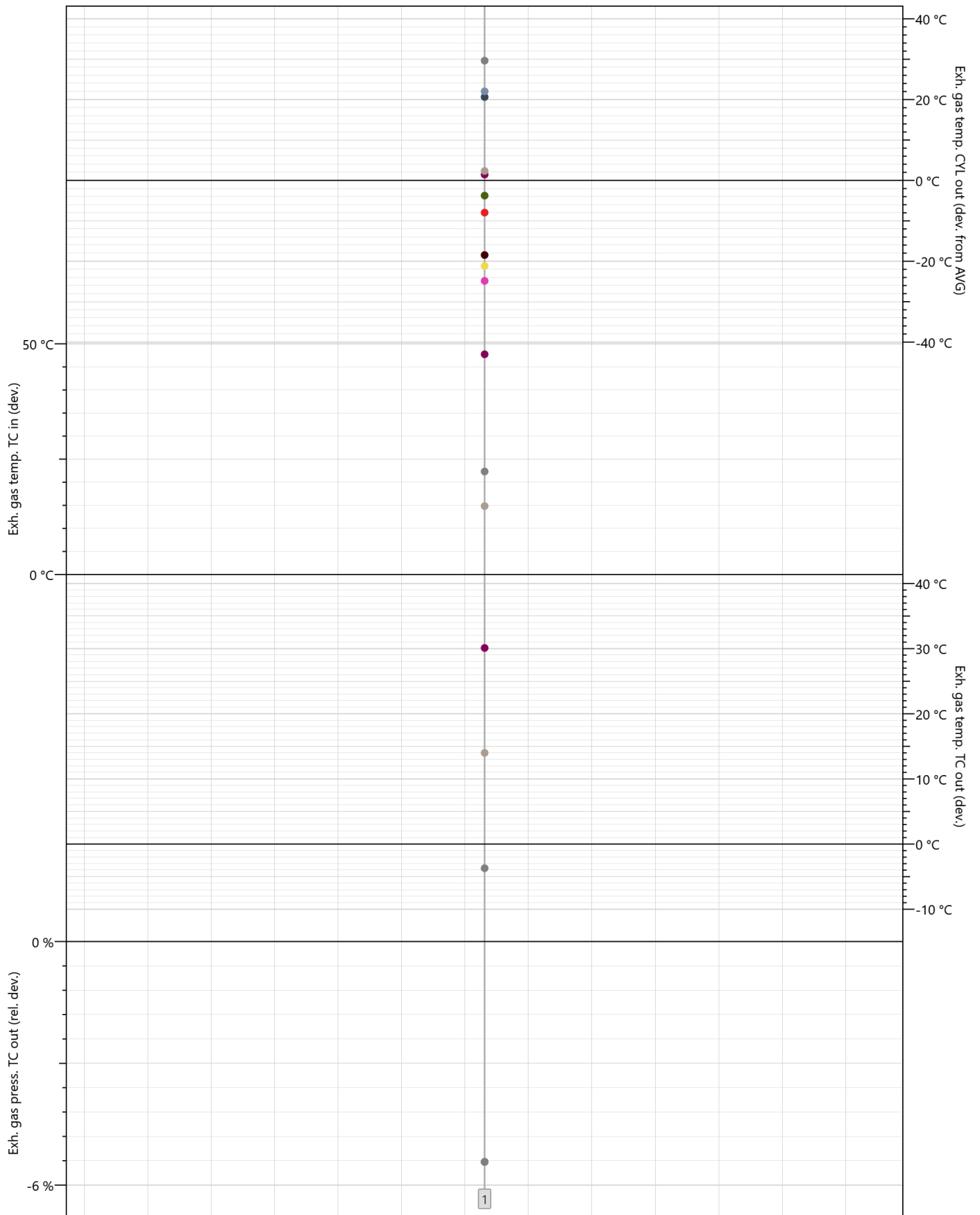
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Trend: EXHAUST SYSTEM

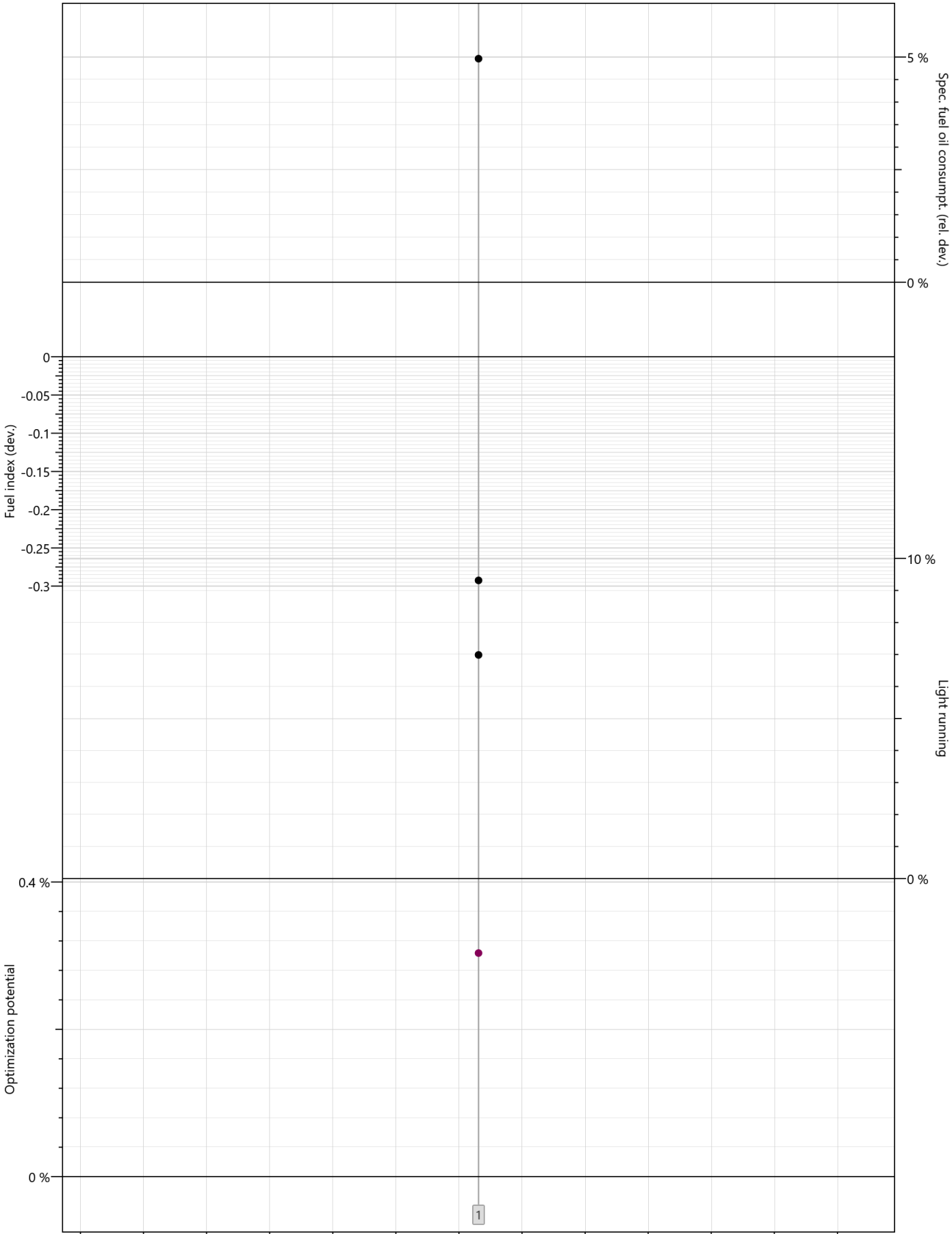
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Trend: CONSUMPTION

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Reading

Installation			Equipment			Type			No.																					
APL HOUSTON			MAIN ENGINE			10S90ME-C			DML0105356																					
General	Date and time of recording		1/24/2026 15:01 : 14		Sea water temp.		23.5 °C		SHIP SOG STW		19.50 18.80 kn		Voy. / title 04FMNW1MA																	
	ENGINE state		stable		Ambient air temp.		25.0 °C		Draft fwd aft		13.70 13.70 m		Record by 2E																	
	ENGINE running hours		70819 h		Engine room air temp.		27.0 °C		Draft mid Trim		13.70 0.00 m		Ackn. by CEW																	
	Ambient humidity		rel 68.0 %		Barom. pr. Eng. room		1.020 1.029 barA		Remarks																					
		abs 13.4 g/kg																												
Power / Speed			CALCULATED		MEAS		ESTIMATED 24.3(E264X126)																							
			XPERT		TORSIOM.		MOP		by FI*rpm		by pscav		by TC rpm		by MEP		by rpm													
	ENGINE power estimated		kW		21439		21630		21449		21296		22004		21368		21021													
			%		42.0				42.0																					
			MCR		ENGINE		TORSIOMETER		MARGINS		MECH. EFFICIENCY																			
ENGINE speed		rpm		84.0 80.1 %		67.3		Power kW 21630		Light running		7.0 %		Theoretic.		93.4 %														
ENGINE power effective		kW		51070 42.0 %		21439				Power		18.3 %																		
(for PTI operation, use negative value, "-")																														
Electronic ctrl	Running mode		Economy		ISO CORRECTED		MEASURED																							
	Governor mode		RPM		REF.		CALC.		ALL																					
	Fuel index ECU		%		59.2		58.9		55.0																					
	Applied fuel quality offset		%				0.0																							
Cylinder pressure	SOURCE: MAN CoCoS		ISO CORRECTED		MEASURED												SETPOINT													
	QUALITY Imported		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10		MOP	
	Firing press. pmax		barG		132.1		132.7		132.7		132.6		132.8		132.6		132.8		132.7		132.8		132.6		132.8		132.7			
	pmax deviation		bar				ALL		0.0		-0.1		0.1		-0.1		0.1		0.0		0.1		-0.1		0.1		0.0			
	pmax offset		bar				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Compression press.		barG		98.0		97.6		97.7		97.8		97.8		97.7		97.7		97.6		97.7		97.8		97.7		97.7			
	pcomp deviation		bar				ALL		0.1		0.1		0.0		0.0		-0.1		0.0		0.1		0.0		0.1		0.0			
	pcomp / pscav offset		-				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Mean indicated press.		barG		9.85		9.68		10.40		9.60		9.60		9.60		9.60		9.60		9.60		9.60		9.60		9.60			
	MIP deviation		bar				0.72		-0.08		-0.08		-0.08		-0.08		-0.08		-0.08		-0.08		-0.08		-0.08		-0.08			
	Fuel offset high		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Fuel offset low		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Power indicated		kW				SUM:		22515		2419		2233		2233		2233		2233		2233		2233		2233		2233			
	Mean effective press. (MEP)		bar		9.22		9.04		51.4 %		of MCR																			
	pmax-pcomp pcomp / pscav		-		34.1 47.9		35.0 46.6												pcomp / pscav											
	Turbocharger			ISO CORRECTED		MEASURED																								
				REF.		CALC.		AVG. / ENG.		TC 1		TC 2		TC 3																
PV (tot-tot)								2.07		2.07		2.07																		
TC speed		rpm		7326		7436		7410		7415		7400		7414																
TC speed 298		rpm		7512		7384		7323		7328		7313		7327																
Turbocharger efficiency																														
Surge margin																														
Air mass flow		kg/s		54.0		53.7		53.5		18.0		17.7		17.8																
Break spec.air cons.		kg/kWh				9.0																								
Exhaust gas mass flow		kg/s		54.5		54.3		54.0		18.1		17.9		18.0																
Scavenge air			ISO CORRECTED		MEASURED																									
	AUX. BLOWER operation		off		REF.		CALC.		AVG.		TC 1		TC 2		TC 3															
	Suction press.		mmWG		18				20		20		20		20															
	Press. drop accr. SAC		mmWG		126				85		90		80		85															
	Scav. air press. RECEIVER		barG		1.07		1.12		1.10																					
	Scav.air press.setp. MOP		barG						1.09																					
	Air temp. COMPR. in		°C						30.0		30.0		30.0		30.0															
	Press. COMPR. outlet *		barG		1.06				1.06		1.06		1.06		1.06															
	Scav. air temp. COMPR. out		°C		115				115		115		115		115															
	Scav. air temp. SAC out		°C						33.0		34.0		34.0																	
	Scav. air temp. RECEIVER		°C		35.9				38.4																					
	Cool. water temp. SAC in		°C		30				32.0		32.0		32.0		32.0															
	Cool. water temp. SAC out		°C						38.0		38.0		38.0		38.0															
	*measured at flange																													
Exhaust gas			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10			
	Exh. gas temp. CYL out		°C		283		282		292		293		294		321		312		314		288		273		271		267		284	
	temp. deviation		°C						ALL		1		2		30		21		22		-4		-18		-21		-25		-8	
	Exh. vlv. open offset		°CA				MOP SETTING:																							
	Exh. gas press. MANIFOLD		barG		0.95		1.02		1.00		TC 1		TC 2		TC 3															
	Exh. gas press. TC in *		barG		0.93				0.98		0.98		0.98		0.98		Check input													
	Exh. gas press. TC out		mmWG		53				50		50		50		50															
	Exh. gas temp. TC in		°C		315		344		353		373		340		348															
	Exh. gas temp. TC out		°C		226		240		249		266		249		232															
	Exh.wastegate (EGB) pos. area %				0 0 0 0																									
	*measured at flange																													

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Installation				Equipment				Type				No.								
APL HOUSTON				MAIN ENGINE				10S90ME-C				DML0105356								
Fuel oil			ISO CORRECTED		MEAS		Fuel oil properties:		ANALYS.		ESTIMATE		Fuel oil system:		REF		MEAS		EXPECT.	
	Absolute consumption	m³/h ▾	REF.	CALC.	4.310		Viscosity	cSt	239		@ 50 °C		Viscosity ENGINE in	cSt	12.5		12.0		14.6	
	Specific FOC	g/kWh	168.2	176.5	184.0		Sulfur content	%	0.49		0.49		Temp. ENGINE in	°C	134		127			
	Light running correction	g/kWh	0.0			Water content	%	0.10		0.10		Temp. FLOWMETER	°C			84				
	Thrust bearing loss	g/kWh	-0.8			LCV	kJ/kg	41290		41290		Density FLOWM.	kg/m³			915.5				
	Specific FOC shoptest	g/kWh	167.4	175.7			Density	kg/m³	959.5		@ 15 °C		Price	USD/t						
							Source of analysis	Laboratory		▾										
Cyl. lub.			REFERENCE		MEAS		Cyl. lubr. system (Alpha):		REF		SET**		Cyl. lub. oil:		REF		ANALYS.			
	Absolute consumption	l/h ▾			20.0		Target feed rate@MCR	g/kWh	0.60		0.70		BN		40 - 70		70			
	Spec. lub oil consumpt. *	g/kWh			0.86		Target f.r.@current load	g/kWh	0.70		SET based		Density	kg/m³			940.0		@ 15 °C	
	* Effective feed rate						** Acc.to OEM operation manuals and/or drainoil analysis							kg/m³			924		@ 40 °C	
Cylinder condition			REFERENCE		MEASURED															
			REF.		AVG.															
			75		ENGINE / CYL 1															
	Cool. water temp. CYL in	°C	75		78.8	79	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10					
	Cool. water temp. CYL out	°C	90		86.2	86	86	86	86	87	86	86	86	86	87					
	Cooling water press. CYL in	bar			5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0					
	Coolant temp. PISTON in	°C			44	44	44	44	44	44	44	44	44	44	44					
	Coolant temp. PISTON out	°C			55	55	54	55	54	55	55	54	54	54	55					
	Scav. air temp. PIST. UNDERS.	°C			44.3	44	41	47	47	46	42	43	45	43	45					
Crankcase			MEASURED																	
			AVG. / ENG.																	
	MAIN BEARING temp.	°C	52.9		52.0	55.5	54.6	54.2	53.9	52.2	49.9	50.7	52.2	54.5	55.4	49.9				
	Temp. deviation	°C			-1	3	2	1	1	-1	-3	-2	-1	2	2	-3				
					CYL 1															
	CRANK PIN BEAR. temp	°C	50.7		50.4	50.8	50.2	50.1	50.8	53.6	51.7	50.3	49.3	49.6						
	Temp. deviation	°C			0	0	0	-1	0	3	1	0	-1	-1						
	X-HEAD BEAR. temp	°C	48.9		47.0	49.0	49.0	50.0	50.0	50.0	48.3	49.0	48.5	48.5						
	Temp. deviation	°C			-2	0	0	0	1	1	1	-1	0	0	0					
Liner wall			MEASURED																	
			CYL 1																	
	LINER WALL temp. aft	°C																		
	LINER WALL temp. fp	°C			125	125	138	134	123	127	117	128	115	126						
	LINER WALL temp. fwd	°C																		
	LINER WALL temp. exh	°C	AVG.		121	126	133	127	132	128	118	127	116	121						
LINER WALL temp. AVG	°C	125																		
TC/SAC media			MEASURED																	
			AVG.				TC 1	TC 2	TC 3											
	Water press. SAC in	bar	2.9		2.9	2.9	2.9													
	Lub oil temp. TC in	°C	45		45	45	45													
Lub oil temp. TC out	°C	60		59	60	59														
Engine media			MEAS		MEASURED															
	Lub oil temp ENGINE in	°C	44.4		Fuel oil press. ENGINE in	bar	7.55													
	Lub oil temp. THRUST SEGM.	°C	56.0		Fuel oil press. ENGINE out	bar	7.30													
	Lub oil press. SYSTEM OIL	bar	2.43		Fuel oil press. FILTER in	bar	7.8													
	Lub oil press. COOLING OIL	bar	2.37		Fuel oil press. FILTER out	bar	7.5													
	Lub oil press. TURB. OIL	bar	2.00		Hydraulic oil press. FILTER in	bar	2.5													
	Lub oil temp. TURB. OIL in	°C	45.00		Hydraulic oil press. FILTER out	bar	2.4													
	Lub oil temp. TURB. OIL out	°C	60.00		Hydraulic oil press. MAIN	bar	228.0													
	Axial vibration	mm	0.83		Sealing air press.	bar			PUMP 2	PUMP 3	PUMP 4	PUMP 5								
					Hydr. oil PUMP swash plate pos.	%	57.0	50.0	57.0											
	Plant			MEAS																
Exh. gas differential press. ECONOMIZER		mmWG	10.00																	